

PHY 615-2012

Problem Set #4

Problem 1 (20.1.1) ($\hbar=c=1$)

$$i \frac{\partial \psi}{\partial t} = -i \vec{\alpha} \cdot \vec{\nabla} \psi + \beta m \psi$$

$$\Rightarrow -i \frac{\partial \psi^\dagger}{\partial t} = i \vec{\nabla} \psi^\dagger \cdot \vec{\alpha} + m \psi^\dagger \beta \quad \left(\begin{array}{l} \alpha^\dagger = \alpha \\ \beta^\dagger = \beta \end{array} \right)$$

Thus

$$\begin{aligned} i \frac{\partial}{\partial t} (\psi^\dagger \psi) &= -i \psi^\dagger \vec{\alpha} \cdot \vec{\nabla} \psi + m \psi^\dagger \beta \psi \\ &\quad - i \vec{\nabla} \psi^\dagger \cdot \vec{\alpha} \psi - m \psi^\dagger \beta \psi \\ &= -i \vec{\nabla} (\psi^\dagger \vec{\alpha} \psi) \end{aligned}$$

$\Rightarrow$

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0 \quad \vec{J} = \psi^\dagger \vec{\alpha} \psi$$

Problem 2 : (20.2.16)

($\hbar=c=1$)

$$\vec{\pi} = \vec{p} - q \vec{A}$$

Need to calculate $\epsilon_{ijk} \pi_j \pi_k f$

$$= \epsilon_{ijk} (-i \partial_j - q A_j) (-i \partial_k - q A_k) f$$

$$= \epsilon_{ijk} \left[-\partial_j \partial_k f + i q A_j \partial_k f + i q \partial_j (A_k f) + q^2 A_j A_k f \right]$$

$$\epsilon_{ijk} \partial_j \partial_k f = 0 \quad \epsilon_{ijk} A_j A_k = 0$$

Thus

$$\epsilon_{ijk} \pi_j \pi_k f$$

$$= iq \epsilon_{ijk} (A_j \partial_k f + (\partial_j A_k) f + A_k \partial_j f)$$

Again

$$\epsilon_{ijk} (A_j \partial_k f + A_k \partial_j f) = 0$$

antisymmetric
under $j \leftrightarrow k$

symmetric under $j \leftrightarrow k$

Thus

$$\epsilon_{ijk} \pi_j \pi_k f$$

$$= iq \epsilon_{ijk} \partial_j A_k f$$

By definition $\epsilon_{ijk} \partial_j A_k = B_i$
 $\Rightarrow$

$$\epsilon_{ijk} \pi_j \pi_k = iq B_i$$

Since f is arbitrary

ie

$$\vec{\pi} \times \vec{\pi} = iq \vec{B}$$

Problem 3 (15.1.2.)

$$(1) \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} [(\vec{S}_1 + \vec{S}_2)^2 - \vec{S}_1^2 - \vec{S}_2^2]$$

Thus the eigenvalues are

$$\frac{1}{2} \hbar^2 [S(S+1) - S_1(S_1+1) - S_2(S_2+1)]$$

For $S_1 = S_2 = \frac{1}{2}$

possible values of $S = 1, 0$

For $S = 1$, the states are triplets and

$$\begin{aligned} E_+ &= -R_y + \frac{A}{2} \hbar^2 [1(1+1) - \frac{1}{2}(\frac{1}{2}+1) - \frac{1}{2}(\frac{1}{2}+1)] \\ &= -R_y + \frac{A}{2} \hbar^2 [2 - 3/2] \\ &= -R_y + \frac{\hbar^2 A}{4} \end{aligned}$$

For $S = 0$ the state is a singlet

$$\begin{aligned} E_- &= -R_y + \frac{A}{2} \hbar^2 [0 - 3/2] \\ &= -R_y - \frac{3}{4} \hbar^2 A \end{aligned}$$

(2) Let ~~g~~ g -factor for the proton be g_p and that of the electron be g_e

H

Then $\mu_e = \frac{g_e e}{2mc}$ $\mu_p = \frac{g_p e}{2Mc}$

$$A\hbar^2 = \frac{g_e e}{2mc} \cdot \frac{g_p e}{2Mc} \cdot \frac{1}{a_0^3} \hbar^2$$

$$= \frac{(g_e g_p)}{4} \frac{(e\hbar c)^2}{mc^2 Mc^2} \frac{1}{a_0^3}$$

Use

$$e^2 = \alpha(\hbar c)$$

$$A\hbar^2 = \frac{g_e g_p}{4} \alpha \frac{(\hbar c)^3}{(mc^2)(Mc^2)} \frac{1}{a_0^3}$$

Use

$$mc^2 = 0.5 \times 10^6 \text{ eV}$$

$$Mc^2 = 940 \times 10^6 \text{ eV}$$

$$\hbar c = (1973) \text{ eV} \cdot \text{\AA}$$

$$g_e = 2$$

$$g_p = 5.6$$

$$a_0 = 0.5 \text{\AA}$$

$$A\hbar^2 \simeq 6 \times 10^{-6} \text{ eV} = \Delta E$$

Then

$$\lambda = \frac{\hbar c}{E} \simeq 21 \text{ cm.}$$

$$(3) \quad \frac{P_{\text{triplet}}}{P_{\text{singlet}}} \sim e^{-\Delta E/kT}$$

Problem 4 (15.2.2)

$$\boxed{\hbar = 1 \text{ UNITS}}$$

Here $j_1 = \frac{1}{2}$ $j_2 = 1 \Rightarrow j = \frac{3}{2}, \frac{1}{2}$
 Consider the top state with
 $j = \frac{3}{2}$.

Clearly

$$|\frac{3}{2}, \frac{3}{2}\rangle = |1, 1; \frac{1}{2}, \frac{1}{2}\rangle$$

To obtain

$$|\frac{3}{2}, \frac{1}{2}\rangle = \text{~~some expression~~}$$

use eqn (15.2-7)

$$|\frac{3}{2}, \frac{1}{2}\rangle = \left(\frac{\frac{1}{2}}{\frac{3}{2}}\right)^{\frac{1}{2}} |\frac{1}{2}(-\frac{1}{2}); 1, 1\rangle + \left(\frac{1}{\frac{3}{2}}\right)^{\frac{1}{2}} |\frac{1}{2} \frac{1}{2}; 1, 0\rangle$$

Thus

$$\langle \frac{1}{2}, -\frac{1}{2}; 1, 1 | \frac{3}{2}, \frac{1}{2} \rangle = \frac{1}{\sqrt{3}}$$

$$\langle \frac{1}{2}, \frac{1}{2}; 1, 0 | \frac{3}{2}, \frac{1}{2} \rangle = \sqrt{\frac{2}{3}}$$

The top state with $j = \frac{1}{2}$ is

$$|\frac{1}{2}, \frac{1}{2}\rangle = +\sqrt{\frac{2}{3}} |\frac{1}{2}(-\frac{1}{2}); 1, 1\rangle - \sqrt{\frac{1}{3}} |\frac{1}{2} \frac{1}{2}; 1, 0\rangle$$

$\Rightarrow$ (with our sign convention)

$$\langle \frac{1}{2}, -\frac{1}{2}; 1, 1 | \frac{1}{2}, \frac{1}{2} \rangle = \sqrt{\frac{2}{3}}$$

$$\langle \frac{1}{2}, \frac{1}{2}; 1, 0 | \frac{1}{2}, \frac{1}{2} \rangle = -\sqrt{\frac{1}{3}}$$

(2) Here $j_1 = 1$ $j_2 = 1$

Thus $j = 2, 1, 0$

$$|2, 2\rangle = |1, 1; 1, 1\rangle$$

Then

$$|2, 1\rangle = \sqrt{\frac{1}{2}} |1, 0; 1, 1\rangle + \sqrt{\frac{1}{2}} |1, 1; 1, 0\rangle$$

$$|2, 0\rangle = \frac{1}{C_{-(2,1)}} J_- |2, 1\rangle$$

$$= \frac{1}{\sqrt{3 \cdot 2}} (J_{1-} + J_{2-}) |2, 1\rangle$$

$$= \frac{1}{\sqrt{6}} \left(\frac{1}{\sqrt{2}} J_{1-} |1, 0; 1, 1\rangle + \frac{1}{\sqrt{2}} J_{1-} |1, 1; 1, 0\rangle \right.$$

$$+ \frac{1}{\sqrt{2}} J_{2-} |1, 0; 1, 1\rangle$$

$$+ \left. \frac{1}{\sqrt{2}} J_{2-} |1, 0; 1, 1\rangle \right)$$

$$= \frac{1}{\sqrt{6}} \left[\frac{1}{\sqrt{2}} \sqrt{1 \cdot 2} |1, -1; 1, 1\rangle \right.$$

$$+ \frac{1}{\sqrt{2}} \sqrt{2} |1, 0; 1, 0\rangle$$

$$+ \frac{1}{\sqrt{2}} \sqrt{2} |1, 0; 1, 0\rangle + \frac{1}{\sqrt{2}} \sqrt{2} |1, 1; 1, -1\rangle \left. \right]$$

(where we used

$$C_{-}(j, m) = \sqrt{(j+m)(j-m+1)}$$

$$C_{+}(j, m) = \sqrt{(j-m)(j+m+1)}$$

Thus

$$|2,0\rangle = \frac{1}{\sqrt{6}} [|1,-1;1,1\rangle + 2|1,0;1,0\rangle + |1,1;1,-1\rangle]$$

The other vectors with $j=2$ may be written down by using

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = (-1)^{j_1+j_2-j} \langle j_1, -m_1; j_2, -m_2 | j, -m \rangle$$

Thus

$$|2,-1\rangle = \frac{1}{\sqrt{2}} (|1,0;1,-1\rangle + |1,-1;1,0\rangle)$$

$$|2,-2\rangle = |1,-1;1,-1\rangle$$

Now consider $j=1$

The state $|1,1\rangle$ must be orthogonal to $|2,1\rangle$. Thus

$$|1,1\rangle = \frac{1}{\sqrt{2}} (\cancel{|1,1;1,0\rangle} - |1,0;1,1\rangle)$$

(using standard convention.)

Applying J_- on this gives

$$|1,0\rangle = \frac{1}{\sqrt{2}} (|1,1;1,-1\rangle - |1,-1;1,1\rangle)$$

$|1,-1\rangle$ can be written down using symmetry, leading to

$$|1, -1\rangle = -\frac{1}{\sqrt{2}} (|1, -1; 1, 0\rangle - |1, 0; 1, -1\rangle)$$

Now consider $j=0$. The state must be orthogonal to both $|2, 0\rangle$ and $|1, 0\rangle$. ~~This gives~~
Write

$$|0, 0\rangle = \alpha |1, -1; 1, 1\rangle + \beta |1, 0; 1, 0\rangle + \gamma |1, 1; 1, -1\rangle.$$

Requiring $\langle 2, 0 | 0, 0 \rangle = 0$ gives

$$\frac{1}{\sqrt{6}} (\alpha + 2\beta + \gamma) = 0$$

Requiring $\langle 2, 0 | 1, 0 \rangle = 0$ gives

$$\frac{1}{\sqrt{2}} (\alpha - \gamma) = 0$$

Thus the state is

$$|0, 0\rangle = N [|1, -1; 1, 1\rangle - |1, 0; 1, 0\rangle + |1, 1; 1, -1\rangle]$$

The normalization is clearly
 $N = \frac{1}{\sqrt{3}}.$

Problem ⁵: (15.2.3)

Clearly $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$

Then

$$\begin{aligned} \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &= (1 \oplus 0) \otimes \frac{1}{2} \\ &= 1 \otimes \frac{1}{2} \oplus 0 \otimes \frac{1}{2} \end{aligned}$$

However

$$1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$$

Thus

$$0 \otimes \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$$

Problem 5: (15.2.3)

Clearly $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$

Then

$$\begin{aligned} \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &= (1 \oplus 0) \otimes \frac{1}{2} \\ &= 1 \otimes \frac{1}{2} \oplus 0 \otimes \frac{1}{2} \end{aligned}$$

However

$$1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$$

Thus

$$0 \otimes \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$$

Problem 6 (15.2.4)

$$\hbar = 1 \text{ UNITS}$$

Consider the action of J^2 on the state $|l + \frac{1}{2}, m\rangle$

$$J^2 |l + \frac{1}{2}, m\rangle = \hbar^2 (l + \frac{1}{2})(l + \frac{3}{2}) |l + \frac{1}{2}, m\rangle$$

Write

$$J^2 = L^2 + S^2 + 2L_z S_z + L_- S_+ + L_+ S_-$$

$$(L^2 + S^2 + 2L_z S_z) |l, m - \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\rangle$$

$$= \left[l(l+1) + \frac{3}{4} + 2(m - \frac{1}{2})\frac{1}{2} \right] |l, m - \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\rangle$$

$$= (l(l+1) + \frac{1}{4} + m) |l, m - \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\rangle$$

Similarly

$$(L^2 + S^2 + 2L_z S_z) |l, m + \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\rangle$$

$$= (l(l+1) + \frac{1}{4} - m) |l, m + \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\rangle$$

$$L_- S_+ |l, m - \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\rangle = 0$$

$$L_+ S_- |l, m - \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\rangle$$

$$= \sqrt{(l - m + \frac{1}{2})(l + m - \frac{1}{2})} |l, m + \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\rangle$$

$$L-S+ | l, m+\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \rangle \\ = \sqrt{(l+m+\frac{1}{2})(l-m-\frac{1}{2})} | l, m-\frac{1}{2}; \frac{1}{2}, \frac{1}{2} \rangle$$

This gives

~~ϕ~~

$$J^2 | l+\frac{1}{2}, m \rangle$$

$$= \left[\alpha \left(l(l+1) + m + \frac{1}{4} \right) + \beta \sqrt{(l+m+\frac{1}{2})(l-m-\frac{1}{2})} \right] \\ | l, m-\frac{1}{2}; \frac{1}{2}, \frac{1}{2} \rangle$$

$$+ \left[\beta \left(l(l+1) + \frac{1}{4} - m \right) + \alpha \sqrt{(l-m+\frac{1}{2})(l+m-\frac{1}{2})} \right] \\ | l, m+\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \rangle$$

Comparing with the expression for $| l+\frac{1}{2}, m \rangle$ and using (1) we therefore get

$$\alpha \left[l(l+1) + m + \frac{1}{4} \right] + \beta \sqrt{(l+m+\frac{1}{2})(l-m-\frac{1}{2})} \\ = (l+\frac{1}{2})(l+\frac{3}{2}) \alpha$$

$$\beta \left[l(l+1) - m + \frac{1}{4} \right] + \alpha \sqrt{(l-m+\frac{1}{2})(l+m-\frac{1}{2})} \\ = (l+\frac{1}{2})(l+\frac{3}{2}) \beta$$

The first equation gives

$$\beta \sqrt{(l+m+1/2)(l-m-1/2)} = \alpha (l-m+1/2)$$

Thus

$$\frac{\beta}{\alpha} = \sqrt{\frac{l-m+1/2}{l+m+1/2}}$$

We could have started with

$J^2 |l-1/2, m\rangle$ and would have
of course got the same result.